

Antibody-oligonucleotide conjugates in cancer therapy: Potential and Promise

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ABSTRACT

X-Drug Conjugates (XDCs) achieve precise therapy through a modular design comprising three key components: targeting vehicle, linker, and therapeutic payload. Among these, Antibody-Drug Conjugate (ADCs) have demonstrated the most remarkable progress in clinical research. By leveraging their precise targeting mechanism and robust clinical data, ADCs have emerged as frontrunner therapeutics in oncology. Building upon this success, Antibody-Oligonucleotide Conjugates (AOCs) represent an innovative evolution—replacing traditional cytotoxic payloads with gene-modulating oligonucleotides, such as small interfering RNA (siRNA) and antisense oligonucleotides (ASO). This paradigm shift transitions the therapeutic focus from "cell killing" to "gene regulation", opening new avenues for cancer treatment. In this review, we provide an overview of the biology and chemistry of AOCs, examine the key components of AOC design (including the antibody, linker and conjugation chemistry, and oligonucleotides, and highlight promising AOC candidates currently in clinical development for cancer treatment while evaluating both the challenges and opportunities. By discussing recent advances in AOCs, we offer valuable insights into future directions for this innovative class of immunoconjugates and their potential to revolutionize targeted cancer therapies.

1. Introduction

Cancer treatment has undergone a dramatic evolution in recent decades, progressing beyond traditional chemotherapy and radiation to highly targeted precision therapies. This transformation has been driven by the development of innovative treatment modalities, including molecular-targeted drugs, monoclonal antibodies, immune checkpoint inhibitors, bispecific antibodies, and chimeric antigen receptor (CAR) T-cell therapy (F. Wang et al., 2024). Among these, antibody-drug conjugates (ADCs) have emerged as an important breakthrough in oncology, with 13 FDA approved ADCs indicated for the treatment of 20 distinct cancer subtypes (Conilh et al., 2023; Filis et al., 2023; He et al., 2024; R. Wang et al., 2025). Most ADC therapies are currently concentrated in hematologic malignancies, with relatively fewer approved for solid tumors (Parit et al., 2024) (J.-H. Li et al., 2024; S. Li et al., 2025).

ADCs consist of monoclonal antibodies that target cancer cell-specific antigens, and a payload, a cytotoxic anticancer drug, is bound via a linker. The therapeutic mechanism of ADCs involves the antibody-mediated specific recognition of tumor-associated antigens, which triggers antigen-dependent internalization for precise drug delivery and enhanced cytotoxicity selectively in malignant cells, while preserving the integrity of healthy tissues (Ahmed et al., 2025; Dumontet et al., 2023). Compared to monoclonal antibody therapies, ADCs maintain the target specificity and long circulation half-life of monoclonal antibodies while significantly enhancing therapeutic efficacy through targeted delivery of cytotoxic payloads, thereby overcome the limitations of monoclonal antibody treatment alone (Shim, 2020). Meanwhile, compared to standalone antitumor agents, ADCs significantly reduce toxicity through antibody-directed precision drug delivery, thereby effectively widening the therapeutic window (Criscitello

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et al., 2021; Fu et al., 2022), (Chau et al., 2019). Early-generation ADCs faced challenges related to immunogenicity, as murine-derived antibodies often elicited human anti-mouse antibodies. This issue has been largely resolved through advances in antibody engineering, which have led to the replacement of murine antibodies with humanized or fully human antibodies (Kumari et al., 2024). Moreover, effective ADC therapy requires the immunoglobulin component to exhibit high affinity and specificity for tumor-associated antigens. However, many of these targets are also expressed at low levels in normal cells or tissues. This overlapping expression can result in off-target toxicity, which has led to the early termination of several clinical trials due to safety concerns (C. Li et al., 2024; Maecker et al., 2023; Saberi et al., 2024; J. Yu et al., 2024). To address this issue, researchers are actively exploring novel targeting strategies that utilize antibody engineering to recognize tumor microenvironment-enriched neopeptides, thereby improving the tumor selectivity and precision of ADCs (Gu et al., 2024; Y. Jin, Schladetsch, et al., 2022).

The clinical success of ADCs has spurred the emergence of an innovative therapeutic approach — the conjugation of monoclonal antibodies with therapeutic oligonucleotides to construct a new generation of "Antibody-Oligonucleotide Conjugates" (AOCs). AOCs represent a class of molecules formed by the covalent linkage of monoclonal antibodies to specific oligonucleotides, including antisense oligonucleotides (ASOs), small interfering RNAs (siRNAs), aptamers, and microRNAs (miRNAs) (Horejs, 2021; Stoddard et al., 2018). AOCs combine the targeting specificity of antibodies with the therapeutic potential of oligonucleotides, enabling the precise delivery of therapeutic nucleic acid to target cells or tissues (Bäumer et al., 2015). Oligonucleotides, typically 13–25 nucleotides in length, can be designed to modulate gene expression through various mechanisms, including gene silencing/-knockdown, gene activation or competitive inhibition of regulatory proteins, as well as to provide immunostimulatory signals (DeFranciscis et al., 2025) (Jiao et al., 2024; M. Liu et al., 2025; H. Zhang et al., 2024). Compared to small-molecule inhibitors or antibodies, oligonucleotides can be specifically designed to target any gene or protein, offering a flexible approach to address the complex molecular landscape of cancer and the tumor microenvironment. Building upon recent clinical successes—including regulatory approvals across cardiovascular, neuromuscular, and central nervous system (CNS) indications, oligonucleotide therapies demonstrate unique pharmacodynamic properties that counteract resistance mechanisms, representing a critical advantage in optimizing cancer treatment durability (Jiang et al., 2024; Kaczmarek et al., 2017; Mukalel et al., 2019). Nucleic acid drugs require precise delivery to diseased tissues or specific cell types, driving the development of various delivery systems (Bian et al., 2025). Currently, the most widely used delivery strategies for oligonucleotide therapeutics include: (1) chemical modifications such as phosphonothioate backbone engineering and GalNAc conjugation, and (2) lipid nanoparticles (LNPs) carrier systems. The iterative optimization of these technologies has significantly improved the pharmacokinetic stability and liver-targeted delivery efficiency of oligonucleotide therapeutics (Kaczmarek et al., 2017). However, following systemic administration, significant bioavailability challenges persist in extrahepatic tissues such as the central nervous system and skeletal muscle, where formidable biological barriers continue to impede the clinical translation of oligonucleotide therapeutics beyond metabolic disease indications (Malecova et al., 2023). To overcome these limitations, AOCs represent a new approach to targeted delivery in oligonucleotide therapeutics. Antibodies offer dual advantages in addressing oligonucleotide delivery challenges: (1) high-affinity epitope targeting specificity and (2) FcRn-mediated trans-endothelial transport, which collectively overcome critical bio-distribution barriers (Kim et al., 2018). These advantages are achieved through two complementary strategies. Specifically, antibody engineering modifications enable precision delivery at tissue- and cellular-level resolution, while leveraging endogenous receptor-mediated endocytic pathways—particularly transferrin

receptor trafficking—significantly enhances the transmembrane delivery efficiency of therapeutic cargo carriers. AOCs strategically address the key limitations of conventional delivery systems, including nonspecific biodistribution, endosomal entrapment, and lysosomal degradation, through the synergistic combination of antibody-mediated targeting systems with oligonucleotide therapeutic payloads and receptor-guided intracellular trafficking mechanisms (Cochran et al., 2024; Zhou et al., 2025). This innovative therapeutic platform enhances the precision of gene expression modulation, creating new possibilities for cancer treatment. By regulating the functional networks of tumor-associated non-coding RNAs (ncRNAs), AOCs enable comprehensive regulation of tumor cell proliferation, apoptosis, and metastatic potential (Chau et al., 2019). By integrating the antigen specificity of monoclonal antibodies with the gene-silencing capability of oligonucleotides, AOCs establishes a programmable platform for precision therapy. This approach leverages antibody-guided targeting for lesion localization while precisely modulating mRNA expression to disrupt pathogenic protein biosynthesis, offering a unique dual therapeutic advantage in both targeted cancer therapy and genetic disease correction (Cuellar et al., 2015; Qian et al., 2023). In this review, we provide an overview of the biology and chemistry of AOCs, examine the key components of AOC design (including antibody, linker and conjugation chemistry, and oligonucleotides), and highlight AOCs currently in development for cancer treatment. We also discuss the challenges and opportunities of AOCs in cancer treatment, along with the future prospects of this emerging class of immunoconjugates.

2. Development of AOCs: structure and function

2.1. Design and structure of AOCs

The components that make up an AOC are an antigen-specific monoclonal antibody (mAb), a stable cleavable or non-cleavable chemical linker, and a potent oligonucleotide payload. When designing an AOC, there are important consideration to take into account.

2.2. Antibodies

In developing of antibody strategies for AOCs, systematic reference can be made to the well-established ADC platform, drawing from approved or clinically investigated ADCs (Colombo et al., 2024; Sasso et al., 2023). Monoclonal antibodies (mAbs) serve as the structural cornerstone of ADCs, fulfilling multiple critical functions (Blay and Pandiella, 2024; K. Liu et al., 2024). Among available antibody formats, IgG mAbs have emerged as the preferred scaffold for ADC development due to their: (1) high-affinity antigen-binding, (2) optimal pharmacokinetic properties (including extended systemic half-life), (3) established large-scale manufacturing processes, and (4) balanced tissue penetration characteristics (De Marco, 2023; Ruan et al., 2024; J. Yu et al., 2020; B. Zhang et al., 2024). Similarly, an ideal antibody scaffold for AOCs should meet several key criteria: (1) high target antigen specificity, (2) efficient induction of cellular internalization, (3) low immunogenicity, and (4) favorable pharmacokinetic stability to prolong systemic circulation time (S. Jin et al., 2022; Qian et al., 2023; Rubahmya et al., 2024). Beyond full-length IgGs, various engineered antibody formats—including Fab fragments, single-chain variable fragments (scFvs), and bispecific antibodies (BsAbs)—have become central to targeted drug delivery systems. These formats offer distinct advantages: Fab fragments and scFvs improve tissue penetration, BsAbs enhance targeting capability, and nanobodies (derived from camelid VHH domains) retain full IgG-like antigen binding while offering unique structural properties (Bannas et al., 2017; Maksymova et al., 2025; Tripathy and Pande, 2025; Y. Wang et al., 2024; Wei and Wei, 2019). These formats are recognized as optimal carriers for oligonucleotide-based therapeutics owing to their precise antigen specificity and sustained

circulatory persistence (Chernikov et al., 2019; Roberts et al., 2020; Yarian et al., 2019).

Notably, key structural parameters from ADC platforms—such as IgG subclass selection, complementarity-determining region (CDR) optimization, target antigen screening criteria (emphasizing tumor specificity and internalization efficiency), and molecular engineering strategies (including Fc modification and stability enhancement)—can be systematically adapted for AOC development (Gao and Sun, 2025; Jiao et al., 2024; Yao et al., 2012). Collectively, these antibody formats and engineering principles provide a strategic framework for rational AOC antibody design (Fig. 1.).

2.3. Linker

Linker design critically governs the core performance metrics of AOCs, including the oligonucleotide to antibody ratios (OARs), therapeutic index, pharmacokinetic behavior, and structural integrity (Nanna et al., 2020; Weng et al., 2019). An ideal linker molecule should exhibit three key characteristics: (1) site-specific conjugation capability, (2) preservation of antibody-binding avidity through structural minimalism, and (3) scalable synthetic accessibility with cost-effective manufacturability (Dugal-Tessier et al., 2021; Jiao et al., 2024). Unlike typical small-molecule drugs, oligonucleotides exhibit broader interaction capabilities. Linkers for AOCs are generally classified into two categories: cleavable and non-cleavable. Fig. 2. illustrates the principal fabrication methods for AOCs.

2.3.1. Non-covalent conjugation strategies

The polyanionic nature of oligonucleotide backbones provides an innate electrostatic docking platform for constructing non-covalent bioconjugates (Cochran et al., 2024). This strategy takes advantage of robust electrostatic interactions between the anionic phosphodiester backbone and arginine-rich protamine peptide motifs to facilitate oriented assembly of antibody-oligonucleotide complexes (Dugal-Tessier et al., 2021). Studies have shown that this prostate-specific membrane antigen (PSMA) antibody-protamine-based delivery system can efficiently load siRNA without chemical modification, significantly inhibiting tumor cell proliferation and lesion progression in murine models of castration-resistant prostate cancer (Mangla et al., 2023; Shi et al., 2019; Ullah et al., 2025). The main benefits of this method are its simplicity and modular flexibility, enabling versatile conjugation of various oligonucleotide sequences to polycationic protein scaffolds.

The biotin-streptavidin system demonstrates unique benefits for

constructing AOCs. By using high-affinity binding between biotinylated oligonucleotides and streptavidin, researchers can achieve precise molecular orientation through programmable directional assembly (Xia et al., 2009). While polycationic complexes are often affected by physiological salt interference, this system better maintains structural stability during circulation. Although it has a stronger binding affinity than protamine-mediated electrostatic assembly, its use is limited because it requires multi-step procedures, including biotinylation pretreatment and gradient purification, making it less practical than conventional conjugation strategies.

Non-covalent conjugation platforms enable modular assembly through the use of standard mAbs and functionalized oligonucleotides, facilitating rapid construction of multifunctional systems while circumventing the antibody activity loss and batch variability associated with chemical crosslinking approaches. Nevertheless, their dynamic binding nature may limit therapeutic applications that demand sustained complex integrity (Jiao et al., 2024; Nanna et al., 2020).

2.3.2. Site-specific conjugation methods

Creating site-specific conjugates in AOCs relies on the chemical properties of specific antibody surface residues. As shown by Thio-Bridge™ conjugation technology, engineered cysteine residues can form bonds through pH-controlled (6.5–7.5) Michael addition with maleimide derivatives (Dugal-Tessier et al., 2021). Lysine more easily forms stable amide bonds with N-hydroxysuccinimide esters or dibenzocyclooctyne when under mildly alkaline conditions, which is the basis of Site-specific Multi-functional Antibody Conjugation technology (Aggarwal et al., 2023). Using cysteine-directed conjugation allows for precise control of OAR values (2–4) control, though it needs strict reducing conditions (Hong et al., 2023). Lysine conjugation permits higher OAR values (8–12) but may cause heterogeneity due to coupling patterns. Recent progress has moved toward dual-conjugation systems, such as combining cysteine site-specific conjugation with enzymatic lysine modification (e.g., microbial transglutaminase-mediated Gln-tag labeling), allowing precise assembly of multiple payloads (Fu et al., 2022). In contrast to ADC developers' strategy of maximizing drug payloads to enhance cytotoxic effects, AOC developers prioritize product consistency and delivery safety, as exemplified by AOC 1001, which features a deliberately constrained OAR of 1 (Benizri et al., 2019; Mul-lard, 2022).

2.3.3. Advances in linker technology

After almost 30 years of technological progress, linker engineering

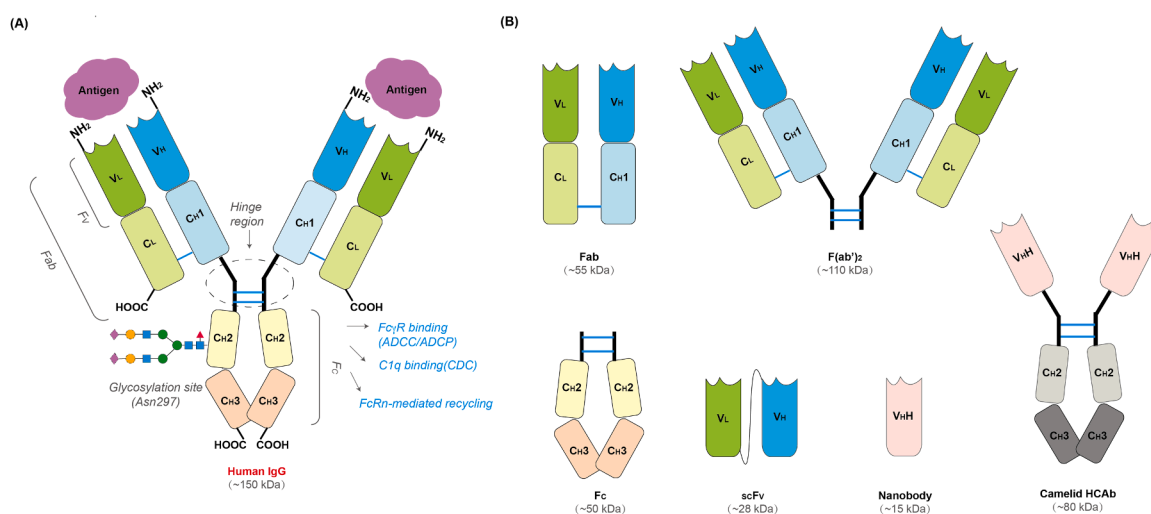


Fig. 1. (A) Structure of IgG antibody and the functional effects. (B) Structures of commonly used therapeutic antibody fragments and mimetics. Fab, antigen-binding fragment; Fc, crystallizable fragment; F(ab)₂, two antigen-binding (Fab) regions linked by disulfide bonds; scFv, single-chain variable fragment; HCAb, heavy-chain-only antibody; Nanobody, a single-domain antibody consisting of a single monomeric variable antibody domain.

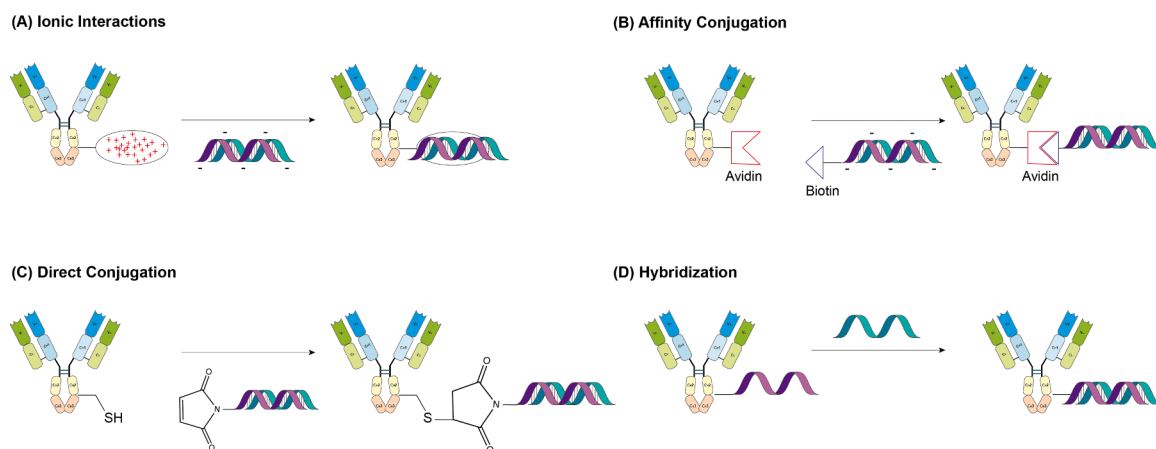


Fig. 2. General methods to conjugate oligonucleotides by (A) electrostatic interactions, (B) biotin and avidin affinity conjugation, (C) direct antibody conjugation and (D) double-stranded hybridization.

for AOCs has developed systematic pathways (Dudchak et al., 2024). Current work focuses on three main areas: (1) making uniform AOCs through site-specific conjugation, such as by using engineered cysteine thiol-ene click chemistry, (2) constructing modular assembly platforms that combine disulfide bonds and pH-sensitive hydrazone bonds for efficient AOCs, and (3) using structural engineering approaches to increase conventional OAR limitations while enhancing payload capacity without compromising product homogeneity (Y. Jin, Edalatian Zakeri, et al., 2022; Wei and Wei, 2019; Zhao et al., 2023). The third-generation smart linkers have significantly improved pharmacokinetic properties via boronic acid-sialic acid dynamic covalent bonds and PEGylated phospholipid bilayers (Park et al., 2016). However, current linker technologies still struggle to balance tumor penetration with response to the tumor environment, leading to new developments like DNA-based connectors and photo-switchable conjugation systems (Kostova et al., 2021).

2.4. Oligonucleotides

Xeno nucleic acids (XNAs) represent chemically engineered nucleic acid analogues that have been utilized in diverse biotechnological applications, including ASOs, siRNAs, and nucleic acid aptamers (Morihiro et al., 2017). Key parameters of oligonucleotides, including length and composition, determine their functionality in structural roles, targeting, detection, gene silencing, and regulation of gene expression (Alshaer et al., 2021; Chernikov et al., 2019). Among all oligonucleotide therapeutics, siRNAs show particular promise due to their catalytic mechanism: each siRNA molecule can silence multiple target RNAs with sequence specificity (Q. Li et al., 2024). These 20–23 nucleotide double-stranded noncoding RNAs induce post-transcriptional gene silencing through a mechanism shared with microRNAs (miRNAs) (Friedrich and Aigner, 2022). siRNAs originate from short hairpin RNAs (shRNAs) or long double-stranded RNAs (dsRNAs), the latter forming when bidirectional transcription produces complementary strands that anneal into dsRNAs (Neumeier and Meister, 2021).

Both ASOs and siRNAs engage mRNA or pre-mRNA targets via complementary base-pairing but exhibit distinct mechanisms (Sabat et al., 2024; Weng et al., 2019). ASOs are heavily modified single-stranded synthetic oligomers that selectively bind target mRNAs via complementary base-pairing, with proven therapeutic efficacy in multiple diseases (Jain et al., 2025). Synthetic ASOs can bind complementary pre-mRNA sequences to alter spliceosomal factor recruitment to modulate splicing outcomes (McKenzie et al., 2021; F. Wang et al., 2022). When bound to mature mRNAs, ASOs either block ribosome engagement, inhibiting protein translation, or recruit RNase H for target transcript degradation (Bege and Borbás, 2022; Márton et al., 2025).

Non-coding RNAs (ncRNAs) play increasingly pivotal roles in detecting, targeting, and treating abnormalities linked to physiological processes, pathological conditions, genetic and epigenetic regulation, cancer, and developmental disorders (Adams et al., 2017). Discussions on personalized cancer medicine increasingly highlight ncRNAs' therapeutic potential. These molecules critically regulate cancer progression and metastasis by modulating oncoprotein and tumor suppressor expression at both transcriptional and post-transcriptional levels (Toden et al., 2021; T. Wang et al., 2017).

2.5. Chemical modification for enhanced payload performance

Chemical modifications of RNA are equally critical for effective nucleic acid delivery, providing both nuclease resistance and immune evasion capabilities (Bramsen et al., 2009; Kaushal, 2023). These modifications, which involve strategic alternations to RNA nucleobases, sugar moieties, or phosphodiester backbones, significantly enhance multiple therapeutic properties, including nuclease resistance, functional efficacy, targeting precision, and cellular delivery efficiency (Pei and Buyanova, 2019). The resulting modified RNA oligonucleotides exhibit improved pharmacological characteristics such as *in vivo* target-binding affinity, greater resistance to degradation, and superior bioavailability (M. Zhang and Huang, 2022).

However, the development of nucleic acid therapeutics continues to face substantial challenges that must be addressed, including delivery system limitations, short circulatory half-life with rapid clearance, problematic off-target binding, poor cellular uptake rates, inefficient endosomal escape, pH sensitivity, nuclease degradation susceptibility, unwanted immunogenicity, and intracellular trafficking barriers (Anand et al., 2025; Hassler et al., 2018). To overcome these limitations, researchers employ multifaceted strategies focusing on siRNAs and targeted delivery optimization, particularly through chemical modifications of nucleotides or backbone linkages that effectively reduce off-target binding and immunostimulation (Ali Zaidi et al., 2023; Kaczmarek et al., 2017). Recent advances in siRNA modification techniques have substantially improved RNA interference technology through various structural enhancements including ribose moiety alterations, nucleobase modifications, phosphodiester backbone adjustments, and incorporation of non-nucleotide entities (Bramsen et al., 2009).

For instance, the incorporation of phosphorothioate linkages along with 2'-O-methyl or 2'-fluoro groups into ribose moieties significantly enhances siRNA stability and target affinity, thereby boosting gene-silencing efficacy (Gangopadhyay and Gore, 2022; Q. Tang and Khvorovova, 2024). When paired with optimized delivery ligands, these modifications promote enhanced siRNA accumulation and therapeutic

effectiveness in target tissues (Hassler et al., 2018). Researchers have successfully developed multiple chemical modification strategies that boost siRNA potency while minimizing off-target effects (Gangopadhyay and Gore, 2022; Kaushal, 2023). Although unmodified siRNA with exposed 2'-OH groups exhibits higher activity in cell culture, engineered partial or complete chemical modifications typically enhance molecular stability while potentially compromising biological activity (F. Wang et al., 2022). The most clinically significant modifications include: 2'-O-methyl group substitutions, 2',4'-constrained nucleic acid designs, and ribose moiety alterations (Wittrup and Lieberman, 2015). These structural modifications provide two critical benefits. Firstly, they effectively block 2'-OH-mediated hydrolysis, providing strong nuclease resistance. Secondly, they preserve RNA-induced silencing complex (RISC) compatibility while significantly reducing immunogenicity (Anand et al., 2025; M. Zhang and Huang, 2022). As shown in Fig. 3., different types of oligonucleotide modifications can be implemented to achieve superior molecular stability, enhanced resistance to nuclease degradation, and reduced non-specific binding to plasma protein.

3. Mechanism of AOCs

AOCs are engineered therapeutics consisting of monoclonal antibody site-specifically linked to sequence-defined oligonucleotide payloads via chemical adapters (Bäumer et al., 2015). After the antibody selectively binds to specific receptors on the surface of target cells, the AOC is internalized. Cellular uptake of AOCs is facilitated by multiple endocytic pathways, including clathrin-mediated endocytosis (CME), caveolae-mediated endocytosis, and clathrin-/caveolae-independent mechanisms (Kalim et al., 2017; Khan and Steeg, 2021; Pei and Buyanova, 2019). Once inside the cell, the oligonucleotide is released to bind to or degrade disease-causing RNA. As shown in Fig. 4. and Fig. 5., antibody-siRNA conjugates and anti-ASO conjugates utilize distinct mechanisms to regulate gene expression (Jiao et al., 2024).

3.1. Mechanism of antibody-siRNA conjugates

siRNA is a double-stranded RNA molecule approximately 20–25 nucleotides in length. It guides the RISC to target and degrade specific mRNA transcripts (Raja et al., 2019). The mechanism involves: (1) RISC activation: After processing by the cellular enzyme Dicer, the short double-stranded RNA is loaded into the RISC. (2) mRNA degradation: The siRNA guides RISC to bind to the complementary sequence in the target mRNA, leading to its cleavage and degradation. (3) Post-transcriptional regulation: By silencing the target mRNA, siRNA reduces translation and downstream protein synthesis (Anand et al., 2025; Lehot et al., 2021).

Cuellar TL et al. utilized TH10MAB™ antibody conjugates targeting cell-surface antigens to investigate the intracellular trafficking and gene-silencing potential of antibody-RNA conjugates (ARCs). The study demonstrated that antigen-specific internalization can occur through distinct pathways, including slow or non-internalizing routes, recycling endosomes, and lysosomal trafficking. Importantly, they identified components of the endolysosomal system as potentially critical facilitators of ARC-mediated functional gene silencing (Cuellar et al., 2015). Consistent gene silencing observed with anti-TENB2 ARCs suggests that only a small fraction of internalized siRNA needs to reach the cytoplasmic RISC complex and undergo strand separation to achieve functional silencing. These findings underscore the importance of efficient endosomal trafficking and escape, highlighting the pivotal role of endolysosomal machinery in supporting the intracellular delivery and efficacy of AOC-based therapeutics (Cuellar et al., 2015; G. Tang, 2005).

3.2. Mechanism of antibody-ASO conjugates

ASOs are short, single-stranded DNA or RNA molecules designed to hybridize with the target mRNA sequence and inhibit gene expression.

The mechanisms include: (1) Transcription inhibition: ASOs bind complementarily to the target mRNA to block transcriptional elongation or processing. (2) RNA degradation: ASOs can recruit RNase H1, an endogenous enzyme that degrades the RNA strand of RNA-DNA hybrids, thereby reducing the mRNA levels and their downstream effects. (3) Translation inhibition: By binding to the 5' untranslated region (5' UTR) or coding sequence of the mRNA, ASOs can sterically block ribosome binding or elongation, thereby decreasing protein synthesis (M. Liu et al., 2025). RNase H1-dependent ASOs utilize endogenous nuclease activity, which is widely expressed across most cell types, to degrade target mRNA transcripts. These ASOs typically incorporate short DNA segments that hybridize with complementary mRNA sequences, forming DNA-RNA duplexes. This structure is recognized by RNase H1, which cleaves the RNA strand, thereby inhibiting translation and producing a therapeutic effect (F. Wang et al., 2022). In contrast, steric-blocking ASOs modulate gene expression via non-degradative pathways. By binding to and masking the AUG start codons within open reading frames (ORFs), these ASOs inhibit translation initiation. Additionally, when directed against upstream open reading frames (uORFs), steric-blocking ASOs can relieve uORF-mediated repression and promote the translation of primary ORFs, thereby enhancing targeting protein expression.

In summary, AOC leverages specific antibody-receptor interactions to selectively deliver therapeutic oligonucleotides to targeted tissues or cell subpopulations that are often inaccessible by conventional delivery strategies. siRNAs and ASOs modulate gene expression through distinct mechanisms: siRNAs harness the RNA interference (RNAi) pathway for targeted mRNA destruction, while ASOs act through RNase-mediated mRNA degradation or steric hindrance. Both modalities represent advanced RNA-targeted therapeutics strategies, each with unique advantages and specific considerations for clinical application (S. Guo et al., 2024; Yang et al., 2022).

However, despite these mechanistic strengths, significant barriers remain to the effective intracellular delivery and functional performance of AOCs. First, electrostatic repulsion between the negatively charged oligonucleotides and the anionic phospholipid bilayer limits passive diffusion across the cell membrane (Dume et al., 2024). Unlike hydrophobic molecules that can readily diffuse across membranes, oligonucleotides—being large, hydrophilic polyanions—require energy-dependent internalization pathways (Pei and Buyanova, 2019; Uddin et al., 2023). Secondly, lysosomal trafficking presents a critical barrier to effective delivery. However, the incorporation of cleavable linkers into AOC designs can help circumvent lysosomal degradation by enabling endosomal escape, thereby significantly enhancing therapeutic efficacy (Gu et al., 2024; Vocelle et al., 2020). In contrast, AOCs with non-cleavable linkers rely on enzymatic hydrolysis by lysosomal proteases, such as cathepsin B and plasmin for payload release (Jiao et al., 2024).

Additional challenges include the potential for off-target effects within cells due to unintended binding to homologous non-target RNAs or proteins. For instance, siRNA-loaded RISCs may tolerate mismatches, inadvertently silencing unintended genes (Leng et al., 2017; Ripoll et al., 2024; Wenson et al., 2024). Although AOCs combine the target specificity of antibodies with the sequence precision of oligonucleotides to mitigate such risks (Grau and Wagner, 2024; Xu, 2015), off-target effects remain a concern—especially when AOCs enter cells via non-specific endocytosis, leading to payload release in non-targeted cells (H. Guo et al., 2024; Khan and Steeg, 2021; Mellman and Yarden, 2013). Cytosolic oligonucleotides can induce promiscuous transcriptional and translational suppression, thereby contributing to cell death. Moreover, AOC efficacy also depends on antigen expression levels. With higher levels correlating with improved therapeutic outcomes. Thus, antigen expression levels and intracellular fate of oligonucleotides are both key determinants of therapeutic outcome.

These challenges collectively highlight the need for continued optimization in AOC design—particularly in refining linker chemistry,

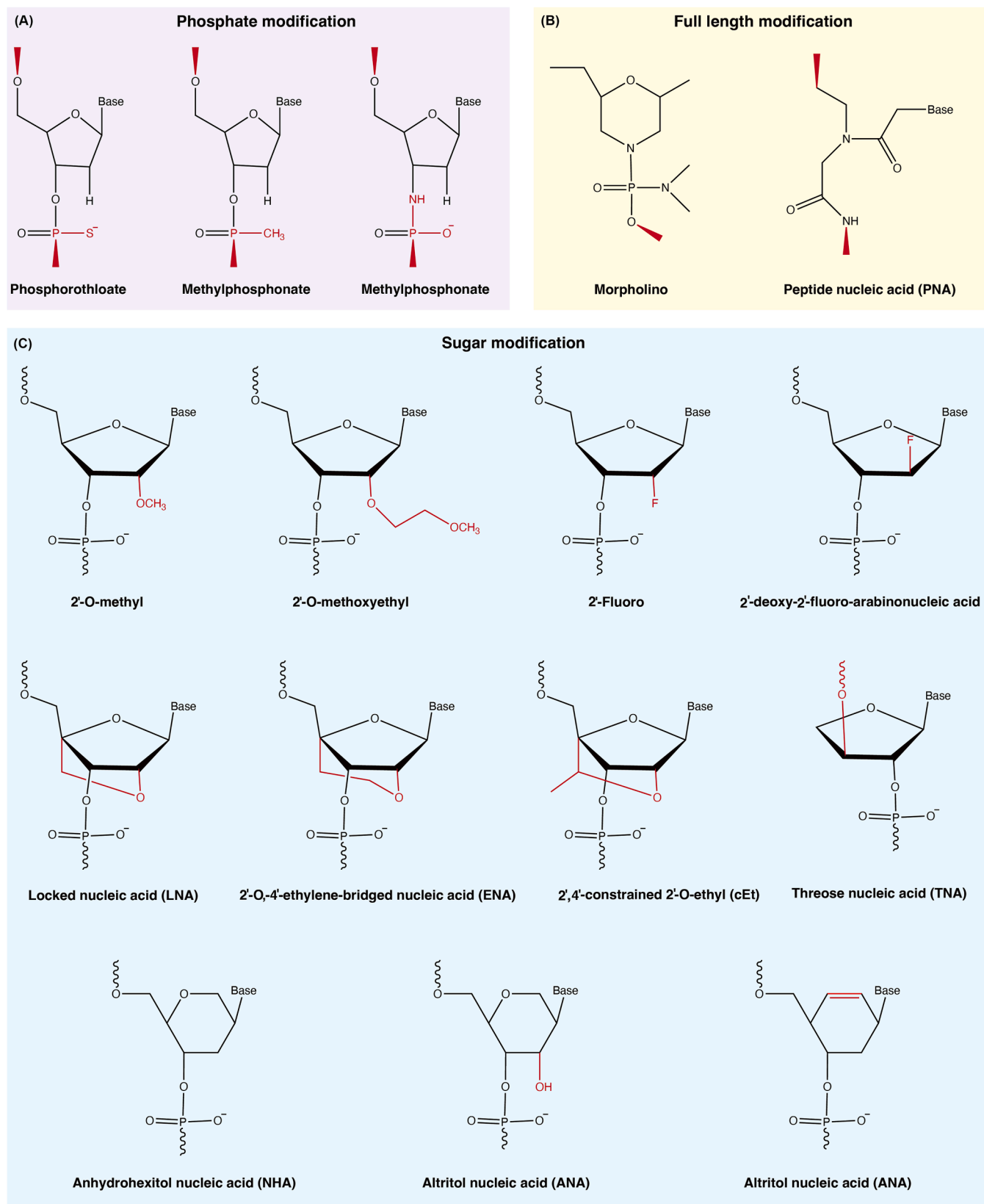


Fig. 3. The chemical modification of oligonucleotides.

Fig. 3. (A) and (B) Phosphate backbone or nucleotide modifications have also been employed to enhance siRNA stability. Specifically, substitution of 2'-O-methyluridine for uridine or replacement of 2'-deoxy-2'-fluorocytidine for cytidine in siRNA sequences confers enhanced stability while minimizing off-target effects. (C) Modification sugar moieties such as 2'-O-methyl ribose or 2'-fluoro ribose have also exhibited enhanced stability and minimum immunogenicity.

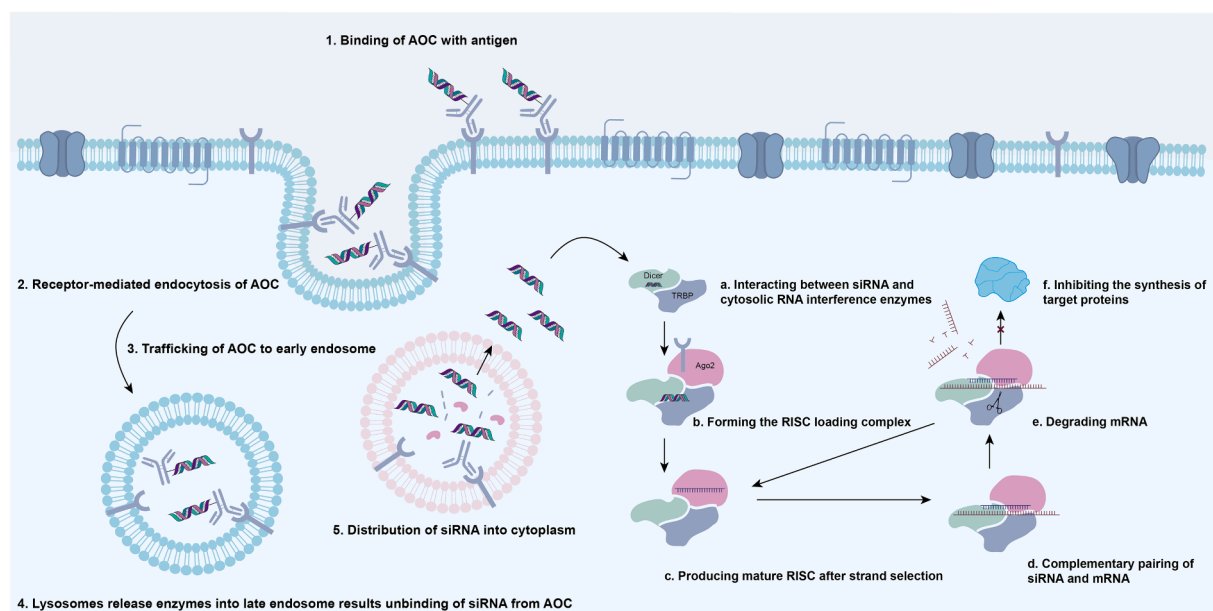


Fig. 4. Key Steps: 1. The antibody portion of the AOC binds to antigens specifically expressed on tumor cells. 2. Triggering internalization instigates early endosomes genesis, which subsequently advance into late endosomes before ultimately merging with lysosomes. 3. The AOC is internalized into vesicles and delivered to early endosomes for processing. 4. Lysosomes release enzymes into late endosome results unbinding of siRNA from AOC. 5. siRNA is released from its carrier into the cytoplasm to reach its functional site. The mechanisms include: (a) siRNA binds to RISC enzymes for gene silencing. (b) siRNA loads into RISC for gene silencing. (c) Functional RISC assembles after selecting its guide strand. (d) Complementary pairing of siRNA and mRNA. (e) The siRNA guides RISC to bind to complementary sequence in the target mRNA, leading to its cleavage and degradation. (f) By silencing the target mRNA, siRNA reduces translation and downstream protein synthesis.

improving endocytic trafficking, and minimizing unintended effects—to fully unlock the clinical potential of this targeted oligonucleotide delivery platform (Ullah et al., 2025; Z. Yu et al., 2021).

4. Current landscape of AOC development

4.1. AOCs enter the clinic

AOCs, initially used as diagnostic tools, include immuno-PCR (iPCR), proximity ligation assay (PLA), proximity extension assay (PEA), electrochemical proximity assay (ECPA), DNA-PAINT imaging, and protein arrays (Dovgan et al., 2019). In recent years, the application of AOCs has expanded beyond diagnostics to encompass targeted therapeutic strategies for a variety of diseases. In 2005, Judy Lieberman from Harvard University laid the groundwork for AOCs by designing the conjugation of siRNA targeting HIV-1 capsid protein mRNA to the Fab fragment using protamine as a linker. By incorporating a fisetin-encoding sequence at the C-terminal end of the antibody heavy chain, this fusion construct enabled the siRNA to form an electrostatic complex with the antibody fragment. As an early prototype, it effectively suppressed HIV-1 expression in infected T cells, representing a pivotal advancement in the early development of AOCs (Song et al., 2005). By leveraging the high binding specificity of antibodies to their target antigens, AOCs allow for the precise delivery of oligonucleotide payloads to tissues or cells that overexpress specific antigens. This targeted strategy enables oligonucleotides to exert their therapeutic effects directly at the disease site, thereby improving treatment efficacy and reducing off-target effects. In October 2021, Avidity Biosciences, a leading innovator in the field of AOC, completed the first patient dosing of AOC 1001 in a Phase I clinical trial—representing the first AOC to reach the clinical trial stage globally. AOC 1001 has since progressed to a phase III trial in May 2024, positioning it as the pioneering candidate in this therapeutic class. In parallel, several pharmaceutical companies are actively advancing their own AOC clinical development programs. An overview of FDA-approved AOCs for clinical use, including the study title, NCT Number, antibody/payload, sponsor, and time, is presented in

Table 1.

Currently, the majority of clinical-stage AOCs are focused on CNS-related indications, particularly neuromuscular disorders such as myotonic dystrophy type 1 (DM1), Duchenne muscular dystrophy (DMD), and facioscapulohumeral muscular dystrophy (FSHD) (Jain et al., 2025). Companies—including Avidity Biosciences and Dyne Therapeutics—have established robust pipelines targeting these conditions. Among these efforts, the transferrin receptor 1 (TfR1) has emerged as a common and critical target. TfR1 is the most extensively studied pathway for transporting antibody-based therapeutics across the blood–brain barrier (BBB), blood–spinal cord barrier (BSCB), and choroid plexus (Hammond et al., 2022). It is predominantly expressed on the luminal surface of brain capillary endothelial cells, where it binds transferrin to facilitate iron transport into the brain parenchyma. AOC 1001 is designed to address the root cause of DM1 by reducing levels of a disease-related mRNA called DMPK in skeletal, cardiac, and smooth muscle. AOC1001 consists of a proprietary mAb that binds to TfR1 conjugated with a siRNA targeting DMPK mRNA. As Avidity’s investigational product candidate for DM1 utilizing its AOC platform is being studied in the Phase III global HARBOR™ trial (NCT06411288) in adolescents and adults living with the disease. AOC1001 is also being studied in the ongoing MARINA-OLE trial (NCT05479981) with all of the participants who completed the Phase I/II MARINA® trial (NCT05027269). Avidity is also developing AOC 144 to treat the underlying cause of DMD. It is designed to deliver phosphorodiamidate morpholino oligomers (PMOs) to skeletal muscle and heart tissue to specifically skip exon 44 of the dystrophin gene to enable dystrophin production in people living with Duchenne muscular dystrophy with mutations amenable to exon 44 skipping (DMD44). AOC 144 consists of a proprietary monoclonal antibody that binds to TfR1 conjugated with a PMO targeting exon 44. The Phase I/II EXPLORE44® trial of AOC 144 has been completed, and the EXPLORE44 Open-Label Extension trial (OLE™) of AOC 144 is currently ongoing. On March 17, 2025, Avidity Biosciences announced topline data from the Phase I/II EXPLORE44® trial, demonstrating highly efficient delivery of PMOs to skeletal muscle, robust increases in dystrophin production, significant enhancement of

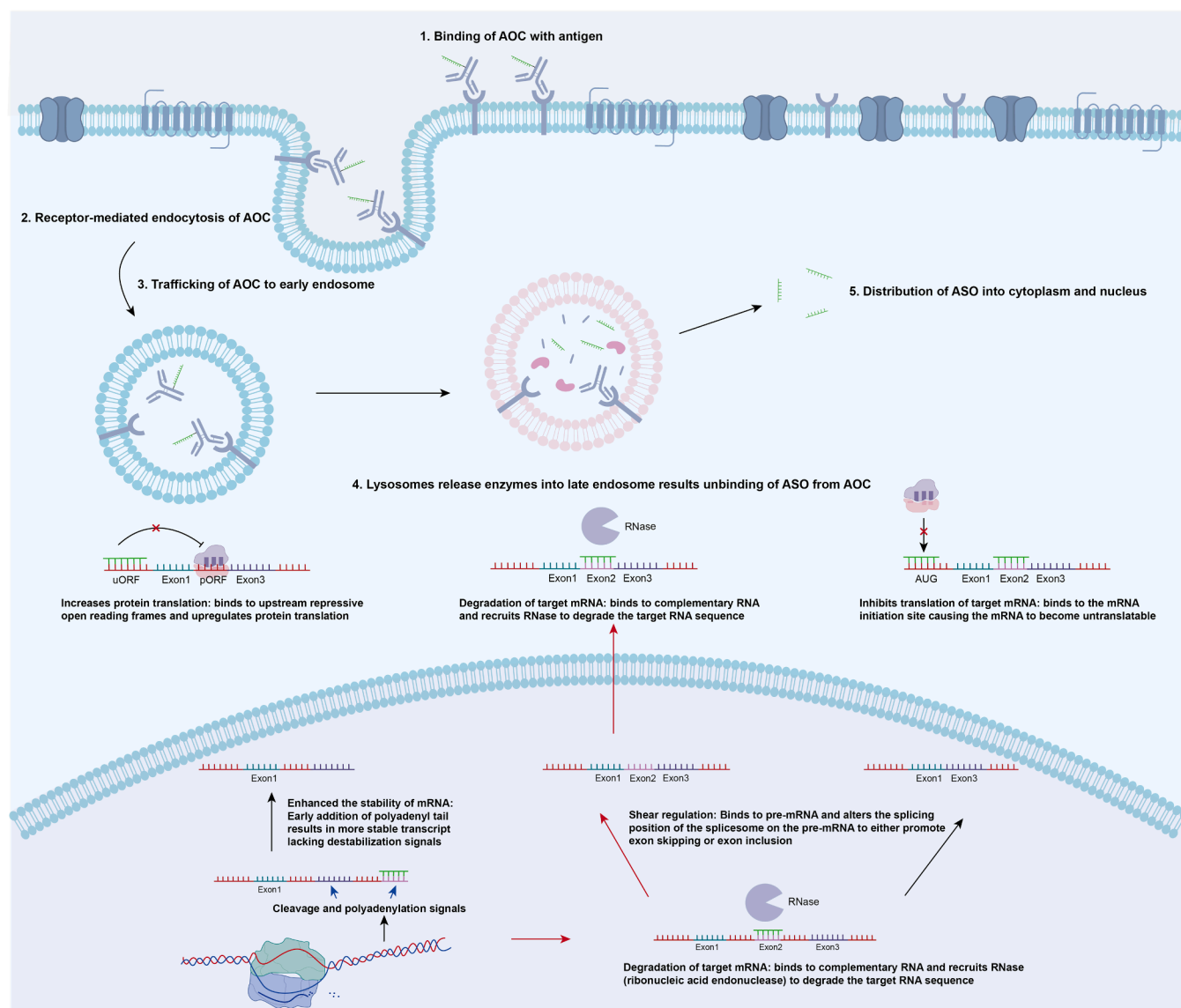


Fig. 5. Key Steps: Steps 1–4 are consistent with the mechanism of antibody-siRNA conjugates. 5. Distribution of ASO into cytoplasm and nucleus. The mechanisms include: **Inhibits translation of target mRNA:** Binds to the mRNA initiation site causing the mRNA to become untranslatable. **Degradation of target mRNA:** Binds to complementary RNA and recruits RNase to degrade the target RNA sequence. **Increases protein translation:** Binds to upstream repressive open reading frames and upregulates protein translation. **Enhanced stability of mRNA:** Early addition of polyadenyl tail results in more stable transcript lacking destabilization signals. **Cleavage and polyadenylation signals:** Guide RNA cleavage and poly-A tail addition during mRNA maturation. **Shear regulation:** Binds to pre-mRNA and alters the splicing position of the spliceosome on the pre-mRNA either promote exon skipping or exon inclusion.

exon 44 skipping, and substantial, sustained decreases in creatine kinase levels to near normal in people living with DMD44 (<https://investors.aviditybiosciences.com/2022-09-27-Avidity-Biosciences-Announces-FDA-Partial-Clinical-Holdon-New-Participant-Enrollment-in-Phase-1-2-MARINA-TM-Trial>). Based on these consistent data, Avidity plans to present functional outcome data in the fourth quarter of 2025 and remains on track to submit a Biologics License Application (BLA) by the end of 2025 for accelerated approval of AOC 144 at a dosing of 5 mg/kg every six weeks in DMD44 (<https://investors.aviditybiosciences.com/2025-03-17-Avidity-Biosciences-Announces-Positive-Topline-Delzota-Data-Demonstrating-Consistent-Statistically-Significant-Improvements-in-Dystrophin,-Exon-Skipping-and-Creatine-Kinase-in-People-Living-with-Duchenne-Muscular-Dystrophy-Amenable-to-Exon-44>). AOC1020 is designed to treat the underlying cause of FSHD, which is caused by the abnormal expression of a gene called double homeobox 4 (DUX4). It consists of a proprietary mAb that binds to Tfr1 conjugated with a siRNA targeting DUX4 mRNA. AOC1020 aims to reduce the

expression of DUX4 mRNA and DUX4 protein in muscles in people with FSHD. Avidity reported positive initial AOC1020 2 mg/kg data at four months from the Phase I/II FORTITUDE™ trial demonstrating unprecedented and consistent reductions of greater than 50 % in DUX4 regulated genes, trends of functional improvement, and favorable safety and tolerability in people living with FSHD (NCT05747924). AOC1020 is currently in Phase II Open-label Extension Study in Participants with FSHD (NCT06547216).

4.2. Promising AOCs in development for cancer therapy

In the oncology field, the only AOC that has progressed into clinical development is TAC-001, developed by Tallac Therapeutics. Tallac developed a novel Toll-like Receptor Agonist Antibody Conjugate (TRAAC) platform to deliver a potent and differentiated Toll-like receptor 9 (TLR9) agonist (T-CpG) for targeted immune activation via systemic administration. TAC-001 is comprised of T-CpG conjugated to

Table 1
Antibody-oligonucleotide conjugates (AOCs) approved by the FDA for clinical use.

Study Title	NCT Number	Antibody	Payload	Status	Sponsor	Time
Global Open-Label Extension Study of Del-desiran for the Treatment of DM1	NCT07008469	TfR1 (mAb)	DMPK siRNA	Phase III	Avidity Biosciences, Inc.	2025.07–2030.10
Global Study of Del-desiran for the Treatment of DM1	NCT06411288	TfR1 (mAb)	DMPK siRNA	Phase III	Avidity Biosciences, Inc.	2024.05–2027.04
Extension of AOC 1001-CS1 (MARINA) Study in Adult Myotonic Dystrophy Type 1 (DM1) Patients	NCT05479981	TfR1 (mAb)	DMPK siRNA	Phase II	Avidity Biosciences, Inc.	2022.08–2027.06
Ph2 Open-label Study of AOC 1044 in Duchenne Muscular Dystrophy Participants With Mutations Amenable to Exon44 Skipping	NCT06244082	TfR1 (mAb)	Exon 44 skipping PMO	Phase II	Avidity Biosciences, Inc.	2024.01–2027.07
Study of AOC 1044 in Healthy Adult Volunteers and Participants with Duchenne Muscular Dystrophy (DMD) Mutations Amenable to Exon 44 Skipping	NCT05670730	TfR1 (mAb)	Exon 44 skipping PMO	Phase I/II	Avidity Biosciences, Inc.	2022.11–2024.11
Phase 2 Open-label Extension Study of AOC 1020 in Participants with Facioscapulohumeral Muscular Dystrophy (FSHD)	NCT06547216	TfR1 (mAb)	DUX4 siRNA	Phase II	Avidity Biosciences, Inc.	2024.07–2028.04
Phase 1/2 Study of AOC 1020 in Participants With Facioscapulohumeral Muscular Dystrophy (FSHD)	NCT05747924	TfR1 (mAb)	DUX4 siRNA	Phase I/II	Avidity Biosciences, Inc.	2023.04–2027.04
Safety, Tolerability, Pharmacodynamic, Efficacy, and Pharmacokinetic Study of DYNE–251 in Participants With Duchenne Muscular Dystrophy Amenable to Exon 51 Skipping	NCT05524883	TfR1 (Fab)	Exon 51 skipping PMO	Phase I/II	Dyne Therapeutics	2022.08–2029.11
Safety, Tolerability, Pharmacodynamic, Efficacy, and Pharmacokinetic Study of DYNE–101 in Participants With Myotonic Dystrophy Type 1	NCT05481879	TfR1 (Fab)	DMPK ASO	Phase I/II	Dyne Therapeutics	2022.09–2029.07
A Dose Escalation and Expansion Study of TAC–001 in Patients With Select Advanced or Metastatic Solid Tumors	NCT05399654	CD22 (mAb)	CpG (TLR9 agonist)	Phase I/II	Tallac Therapeutics	2022.06–2026.05

an anti-CD22 antibody, a receptor restricted to B cells. It is designed to systemically deliver T-CpG to B cells by binding CD22 receptor on B cells, leading to TLR9 signaling, B cell activation and a cascade of immune reactions (Kuo et al., 2021). In November 2023, Tallac reported the first-in-human Phase I clinical trial data, demonstrating that single-agent TAC-001 (0.1–3 mg/kg) given systemically every 2 weeks is well tolerated, demonstrates pharmacodynamic activity consistent with

its proposed mechanism of action and resulted in preliminary clinical activity with patients achieving durable stable disease (≥ 6 months) and partial response per RECIST v1.1 (NCT05399654). Other companies, such as Gennao Bio, are also pursuing AOC-based approaches in oncology. Gennao Bio has developed a proprietary non-viral Genetically Modified Antibody Body (GABA) platform, which features anti-DNA antibodies that employ novel cell-penetrating antibodies to specifically

Table 2
Antibody-oligonucleotide conjugates (AOCs) for cancer therapies.

Antibody	Linker	Oligonucleotide	Indications	Year
HIV envelope (Fab), HER2 (scFv)	Ab-protamine fusion protein	c-Myc, MDM2, VEGF (siRNA)	HIV, Breast cancer	2005 (Song et al., 2005)
LFA–1 (scFv)	Ab-protamine fusion protein	Ku70, cyclin D1, CCR5 (siRNA)	Leukemia	2007 (Peer et al., 2007)
Lewis-Y (mAb)	Ab-HyNic+ 4FB-siRNA (Solulink)	STAT3 (siRNA)	Breast and lung cancer	2011 (Ma et al., 2011)
HER2 (scFv)	Ab-protamine fusion protein	PLK1 (siRNA)	Breast cancer	2012 (Yao et al., 2012)
PSMA (scFv)	Ab-protamine fusion protein	Notch1 (siRNA)	Prostate cancer	2013 (Su et al., 2013)
TENB2, EtBR, MUC6, Mesothelin, NaPi2b, Steap1 (mAb), Her2 (mAb, Her2-EGF antibodies, Fab)	THIOMAB	PPIB (siRNA)	Different cancers	2014 (Cuellar et al., 2015)
EGFR (mAb)	Ab-protamine fusion protein	KRAS (siRNA)	Colon cancer	2015 (Bäumer et al., 2015)
CD22 (mAb)	cyclooctyne-azide	MXD3 (ASO)	Leukemia	2016 (Satake et al., 2016)
Recombinant 2C10 VH	streptavidin-biotin interaction	tNASP (siRNA)	Different cancers	2017 (Im et al., 2017)
CD44, EPHA2 (mAb)	azide-alkyne	DRR/FAM107A (ASO)	Glioblastoma	2018 (Arnold et al., 2018)
PSMA (mAb)	Ab-protamine fusion protein	TRIM24 (siRNA)	Prostate cancer	2019 (Shi et al., 2019)
CD138, BCMA, SLAMF7 (DVD-IgG, DVD-Fab)	lysine- β -lactam	CTNNB1 (siRNA)	Multiple myeloma cells	2020 (Nanna et al., 2020)
CD63 (mAb)	Ionic interactions (9-mer of D-arginine)	miRNA 21	Oral squamous cell carcinoma	2020 (Jiao et al., 2024)
ERFG (mAb)	Cationized gelatin (cGel)	SIKRASG12C (siRNA)	Non-small cell lung cancer	2021 (Sreedurgalakshmi et al., 2021)
EGFR (mAb)	Multifunctional peptides: cell-penetrating peptide (CPP) and substrate peptide (SP)	EGFP (siRNA)	Colon cancer	2021 (Z. Yu et al., 2021)
ERFG Nanobodies (VHH)	LC-SMCC, SM (PEG) 24, APN, LC-SPDP	AHSA1 (siRNA)	Epidermoid cancer, Breast cancer	2021 (Zavoiura et al., 2021)
PD-L1 (mAb)	photocleavable onitrophenyl linker	PD-L1 (siRNA)	Colon cancer	2022 (X. Wang et al., 2022)
CD63 (mAb)	Ionic interactions (linear arginine (9mer), Branched arginine (9 + 9mer))	MYC, CTNNB1 (siRNA)	Multiple myeloma (Exosomes)	2022 (Soma et al., 2022)

deliver nucleic acid payloads to target cells while effectively bypassing endocytic pathways. GMAB-7001, developed using GABA platform as the world's first linker-free AOC, has demonstrated sustained increases in tumor-infiltrating lymphocytes in pancreatic cancer mouse models. It has also shown the ability to cross the blood-brain barrier, reduce intracranial tumor burden by 50 % and prevent spinal metastasis (Mullard, 2022). Although AOCs for cancer therapies have long attracted industry interest, several investigational candidates remain in pre-clinical development or at the proof-of-concept stage for various hematological malignancies and solid tumors. Table 2 summarizes published studies on anti-tumor AOCs to date.

In the development of AOCs for cancer therapy, siRNA and ASO can be conjugated to antibodies either through covalent linkage to engineered lysine or cysteine residues, or via non-covalent interactions with antibody fusion constructs. AOCs represent a promising therapeutic platform due to their precise targeting, improved efficacy, potential to address diverse malignancies, and suitability for personalized medicine paradigms. By combining two emerging therapeutic modalities—nucleic acid drugs and immunotherapy—AOCs enable targeted delivery of therapeutic payloads to malignant cells while minimizing off-target toxicity. However, this dual-modality approach also presents complex challenges in terms of ADME (absorption, distribution, metabolism, excretion), efficacy, and safety, with limited clinical data available to guide development and optimization (Mullard, 2022). The pharmacokinetics of AOCs remain complex, necessitating extensive computational modeling, experimental validation, and clinical evaluation. In addition, clinical translation of AOCs in oncology still faces hurdles, including organ- and cell-selective delivery, intracellular uptake efficiency, endosomal escape, and mitigation of off-target effects at the cellular and molecular levels.

4.3. Limitations and challenges of AOCs in cancer therapy

AOCs represent a novel class of synthetic chimeric biomolecules that has garnered increasing attention across diverse biomedical fields (Dovgan et al., 2019). As a potential successor to ADCs, AOCs can benefit significantly from the extensive experience accumulated in molecular optimization, manufacturing process, and clinical development of ADCs (Kumari et al., 2024; Zhou et al., 2024). Meanwhile, by leveraging the high-specificity targeting capabilities of antibodies and their functional fragments—combined with the modular design capacity for high-affinity composite targets—AOCs offer a promising approach for programmable gene regulation in cancer therapy through precise recognition of malignant cells and their microenvironmental biomarkers (Loganathan and Doss C, 2023; Rady et al., 2024). With advances in elucidating dynamic regulatory mechanisms of tumor signaling networks and the establishment comprehensive molecular target landscapes, AOC technology is poised to transcend current therapeutic boundaries and gain accelerated momentum in clinical applications across both solid tumors and hematologic malignancies.

However, several challenges still impede their clinical therapeutic translation. Firstly, to ensure efficient intracellular trafficking and cytosolic release of oligonucleotide payloads, which is particularly crucial for treating genetically driven cancers, future AOC development must prioritize site-specific conjugation techniques and intelligent delivery systems. Secondly, many preclinical *in vivo* questions regarding AOCs remain unanswered, particularly around the pharmacokinetics (PK) and *in vivo* stability of oligonucleotides, as well as toxicology data in animal models, such as mice, rats, and cynomolgus monkeys. Extensive preclinical validation will be required before clinical translation is feasible. Thirdly, the most promising signs of clinical efficacy to date have come from candidates administered via direct intratumoral injection, which limits their utility to patients with accessible, injectable tumors. Finally, batch-to-batch conjugation heterogeneity in AOC manufacturing remains a critical barrier to clinical advancement. This limitation stands in contrast to the commonly held assumption that

manufacturing complexity has already been adequately addressed.

Emerging research is actively driving innovation in AOC platforms, with focused efforts on reducing immunogenicity, enhancing linker stability, improving cytosolic translocation, identifying more potent and selective oligonucleotide payloads, validating tumor-specific monoclonal antibodies, discovering novel tumor antigens, and optimizing formulations for clinical use. As the field continuously evolves, these advances are likely to pave the way for translational treatments in cancer (Park et al., 2016).

5. Conclusions

With ongoing innovations in antibody engineering, synthetic chemistry, and formulation science, AOCs are poised to become a clinically validated platform for precision therapy and beyond. These advancements are steadily broadening the therapeutic landscape of AOCs, particularly in oncology, immunotherapy, and potentially other disease areas. The successful development of AOC therapeutics relies on the interdependent optimization of three critical components: therapeutic modality, delivery mechanism, and administration route. Although AOCs hold significant promise in cancer treatment, the majority of candidates remain at the conceptual or preclinical stages. Their clinical translation continues to face persistent challenges, including unresolved technical bottlenecks and limited translation efficiency, which demand systematic resolution. Moreover, critical parameters such as synthetic accessibility, conjugate stability, and production efficiency also play decisive roles in determining the success of clinical-stage candidates. Therefore, the long-term clinical value of AOCs still requires rigorous validation through comprehensive investigation and translational studies. In summary, while AOCs offer a novel and promising strategy for targeted cancer therapy, considerable technical and scientific efforts are still needed before their full therapeutic potential can be realized in clinical settings.

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Qinghe Meng: Writing – original draft, Writing – review & editing, Resources. **Mo Yang:** Writing – review & editing. **Fei Xing:** Writing – review & editing. **Zhenxia Xie:** Writing – review & editing. **Ping Jiang:** Writing – review & editing. **Yimeng Hao:** Writing – review & editing. **Baiquan Xiao:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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